



23CSE 452 Business Analytics
Individual Case Study Report

**Predicting Formula 1 Tyre Degradation and Lap-Time
Performance**

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1. Problem Statement and Objectives

Formula 1 is one of the most data-intensive sports in the world. Every race involves hundreds of decisions, and tyre strategy is one of the most important. Tyres wear out as a car drives lap after lap, and as grip drops, lap times get slower. This is called tyre degradation. Teams have to decide when exactly to call a driver into the pit stop for fresh tyres, because pitting too early wastes time and pitting too late causes the driver to lose significant pace on worn rubber.

The challenge is that different tyre compounds — Soft, Medium, and Hard — degrade at different speeds, and this also varies from circuit to circuit. A decision that works at Silverstone might be completely wrong at Singapore. Teams currently rely on experience and real-time radio data from drivers, but these are subjective. A data-driven model that predicts lap time from tyre age and other measurable features would give race engineers a more reliable tool for strategy decisions.

Business problem: Can a machine learning model trained on publicly available lap timing and tyre data predict lap duration accurately enough to support pit-stop strategy decisions, and which features drive tyre-related lap time loss?

The study has three objectives:

- To build an original dataset of 2023 F1 race laps by collecting data from the OpenF1 public API, covering all 22 race sessions of the season.
- To predict lap duration using regression models that take tyre age, compound type, sector times, and other race-context features as inputs, and to compare their performance against a naive session-median baseline.
- To identify which features most drive tyre degradation and convert the findings into practical strategy recommendations for race engineers.

2. Data Collection and Dataset Description

2.1 Source and collection procedure

The data needed for this study was not available in one place, so it had to be scraped from multiple websites and then combined. The two main sources used were the OpenF1 data portal (<https://openf1.org>) (It has API request rate limit for free version and doesn't require any API key or authentication key), the official Formula 1 results website (<https://www.formula1.com/en/results>), and McLaren website (<https://en.mclarenf1.com/>) these websites are freely accessible and do not require any login. Python scripts were written to scrape three separate data streams from these websites across all 22 race sessions of the 2023 season. The three streams were then merged together into a single dataset for analysis. Table 1 summarises what was collected and from where.

Item	Detail
Primary source	OpenF1 data portal (https://openf1.org) — publicly accessible, no login required; structured JSON pages for laps, stints, and driver data per race session [1]
Secondary source	Formula 1 official results website (https://www.formula1.com/en/results) — used to verify round numbers, race names, and session calendar order across the 2023 season. McLaren website (https://en.mclarenf1.com/) – used to collect the lap-by-lap data of driver's sector timings in a lap, tyre compound, pit lap, etc.

Coverage	All 22 race sessions of the 2023 Formula 1 World Championship, from Bahrain (March) to Abu Dhabi (November)
Scraping tool	Custom Python scripts using the requests library to fetch JSON pages; BeautifulSoup and Selenium used for HTML parsing on the F1 results website; responses parsed into pandas DataFrames
Stream 1 — Laps	Lap-level timing: lap number, lap duration, sector 1/2/3 split times, speed trap readings, pit-in and pit-out flags. Raw: 24,479 records, 17 fields
Stream 2 — Stints	Tyre information per stint: compound type (Soft/Medium/Hard), stint number, first and last lap of each stint, tyre age at stint start. Raw: 1,350 records, 9 fields
Stream 3 — Drivers	Driver metadata: broadcast name, full name, car number, team name. Raw: 439 records, 13 fields
Final merged dataset	24,604 rows across 24 columns after joining all three streams on session key and driver number

Table 1: Data collection and web scraping summary.

2.2 Dataset description

The merged dataset contains 24,604 lap records across 24 attributes in three groups. The timing group covers lap duration, sector times (s1, s2, s3), and speed trap readings. The tyre group covers compound name, current tyre age, and stint number. The context group covers session key, driver number, lap number, team name, and pit-in and pit-out flags. The target variable for all models is lap_duration in seconds. Table 2 shows a sample of raw records after cleaning.

Session	Driver	Lap	Compound	Tyre Age	Lap Duration (s)	S1 (s)	S2 (s)	S3 (s)
Bahrain	VER	12	MEDIUM	10	91.34	28.61	40.12	22.61
Jeddah	HAM	8	SOFT	6	90.11	27.93	39.55	22.63
Melbourne	LEC	24	HARD	21	82.47	25.11	36.42	20.94
Miami	ALO	3	MEDIUM	2	91.88	28.12	40.77	22.99
Monaco	SAI	55	HARD	40	75.62	21.77	34.61	19.24

Table 2: Sample records from the raw merged dataset.

#	Column Name	Data Type	Category	Description
1	meeting_key	int64	Race Context	Unique identifier for each Grand Prix race weekend.
2	session_key	int64	Race Context	Unique identifier for each official race session across the 2023 season.
3	driver_number	int64	Driver Info	Permanent car number assigned to the driver (e.g., 1 for Verstappen, 44 for Hamilton).

#	Column Name	Data Type	Category	Description
4	name_acronym	object	Driver Info	Official three-letter driver broadcast acronym (e.g., VER, HAM, LEC).
5	broadcast_name	object	Driver Info	Full driver name as shown in the official timing screen (e.g., M VERSTAPPEN).
6	team_name	object	Driver Info	Formula 1 constructor/team name (e.g., Red Bull Racing, Ferrari, Mercedes).
7	lap_number	int64	Lap Context	Sequential lap number completed by the car in the race (1, 2, 3, ...).
8	date_start	datetime	Lap Context	UTC timestamp indicating the exact start time of the lap.
9	lap_duration	float64	Target Variable	Total lap time in seconds (Target variable for regression models).
10	duration_sector_1	float64	Sector Timing	Split time recorded for track Sector 1 in seconds.
11	duration_sector_2	float64	Sector Timing	Split time recorded for track Sector 2 in seconds.
12	duration_sector_3	float64	Sector Timing	Split time recorded for track Sector 3 in seconds.
13	i1_speed	float64	Speed Trap	Car speed measured at the intermediate speed trap 1 (km/h).
14	i2_speed	float64	Speed Trap	Car speed measured at the intermediate speed trap 2 (km/h).
15	st_speed	float64	Speed Trap	Car top speed recorded at the longest straight-line speed trap (km/h).
16	compound	object	Tyre & Stint	Dry tyre compound type used during the lap (SOFT, MEDIUM, or HARD).
17	stint_number	float64	Tyre & Stint	Current stint index for the driver in the race (Stint 1, 2, 3, ...).

#	Column Name	Data Type	Category	Description
18	lap_start_stint	float64	Tyre & Stint	Race lap number when the current tyre set was fitted during a pit stop.
19	lap_end_stint	float64	Tyre & Stint	Race lap number when the current tyre set was removed.
20	tyre_age_at_start	float64	Tyre & Stint	Initial age of the tyre set at the beginning of the stint (0 for scrubbed/new).
21	tyre_age	float64	Tyre & Stint	Cumulative number of laps driven on the current tyre set (tyre_age_at_start + laps in stint).
22	is_pit_out_lap	bool	Flag	Boolean indicator whether the lap was an out-lap from the pits (False across all retained clean rows).

Table 4: Description of the 22 features and metadata columns in the cleaned Formula 1 lap dataset ($n = 22,733$).

3. Data Preparation and Exploratory Analysis

3.1 Cleaning and preprocessing

Raw race timing data contains many laps that are not useful for predicting normal race pace. Three categories of records were removed before any analysis: laps where the driver is entering or leaving the pit lane (intentionally slow), laps driven on wet or intermediate compounds in rain (a completely different grip condition), and statistical outliers likely caused by safety cars or red flags. Records with missing compound or timing information were also dropped. Table 3 lists each step with the number of records removed.

Removal step	Reason	Removed	Remaining
Missing compound match	Tyre compound not joinable from stint data	100	24,504
Missing lap duration	Lap timing not recorded by API	20	24,484
Pit-out and pit-in laps	Car enters/exits pit lane at non-race speed	923	23,561
Wet / intermediate laps	Different grip physics; not representative	704	22,857
Extreme outlier laps	Above 99.5th percentile per session (SC, flags)	124	22,733

Table 3: Cleaning steps applied to the raw merged dataset.

After cleaning, the dataset contains 22,733 records. A further 428 rows are dropped when rolling features are computed — the first 3 laps of each driver-stint have no rolling history — giving a final modelling dataset of 22,305 laps.

3.2 Feature engineering

Nineteen engineered features were created from the raw numeric columns available after cleaning. The most important additions are: polynomial tyre-age terms (tyre_age squared and cubed) to capture the non-linear acceleration in degradation during the second half of a stint; a 3-lap rolling mean of lap duration per driver per session (rolling_mean_3) to represent the car's current pace level; a per-lap delta from the rolling mean (lap_delta) to capture sudden pace shifts; and a normalised stint progress variable (stint_pct, from 0 to 1) so the model understands where in the stint a lap falls. A compound-age interaction term (compound_x_age) was also added. The final feature matrix used for modelling has 22 columns.

3.3 Exploratory data analysis

Three findings from exploratory analysis shaped the modelling choices. First, tyre degradation is clearly non-linear. Lap times are relatively flat for the first 8 to 10 laps on a fresh set, then rise at an accelerating rate as the rubber reaches its thermal limit. This rules out simple linear models as the primary approach. Second, the compound matters: Soft tyres show the steepest degradation curve, Medium tyres follow a similar but slower curve, and Hard tyres are the most durable with a gentler slope. Third, rolling_mean_3 and the three sector times are the strongest correlates of current lap duration, confirming that recent pace and track-segment context carry more information than static features like driver identity.

Figure 1 shows the lap duration distribution and compound breakdown across the 2023 season.

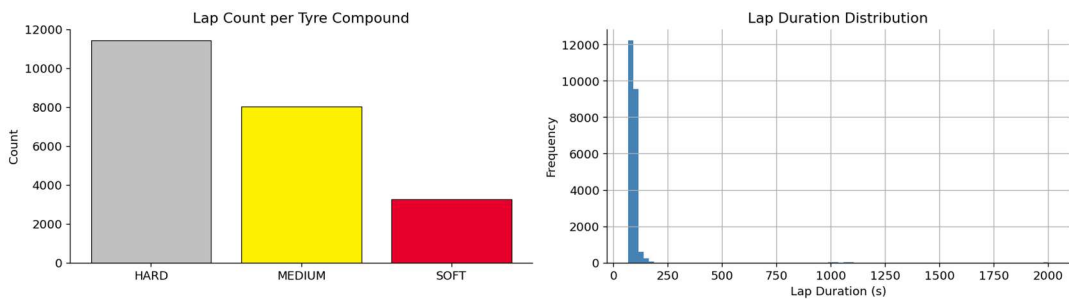


Figure 1: (a) Lap count per compound; (b) Lap duration distribution across the 2023 season.

Figure 2 shows lap duration spread by compound through box plots. Soft tyres have the tightest range over short stints, while Medium and Hard tyres show more spread as they are run for longer.

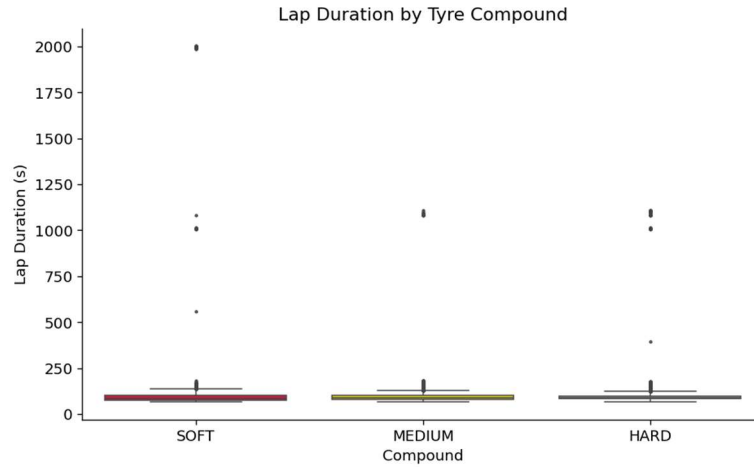


Figure 2: Lap duration by compound (box plot).

Figure 3 shows tyre age plotted against lap duration for each compound separately. The rising median lap time with increasing tyre age confirms non-linear degradation across all three compounds.

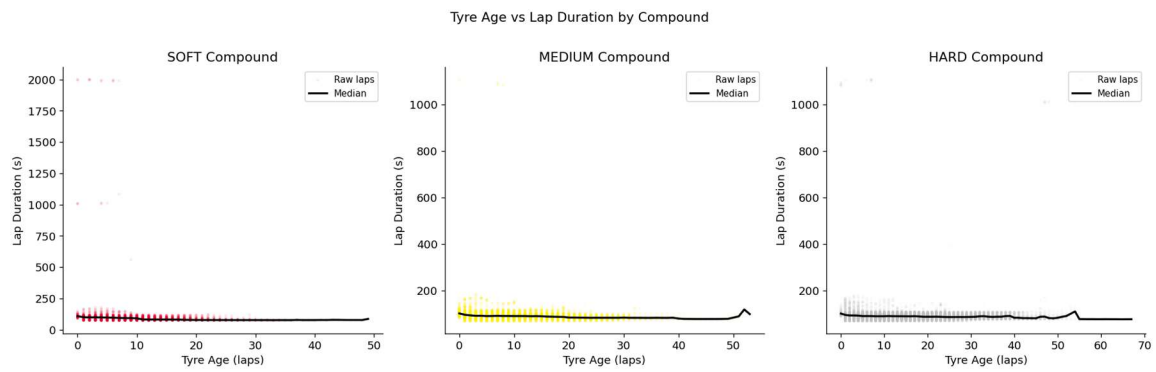


Figure 3: Tyre age vs lap duration by compound — raw laps and median trend.

Figure 4 shows the feature correlation heatmap. Sector times show moderate positive correlation with lap duration, while speed trap readings show negative correlation, which makes sense as faster trap speeds correspond to shorter laps.

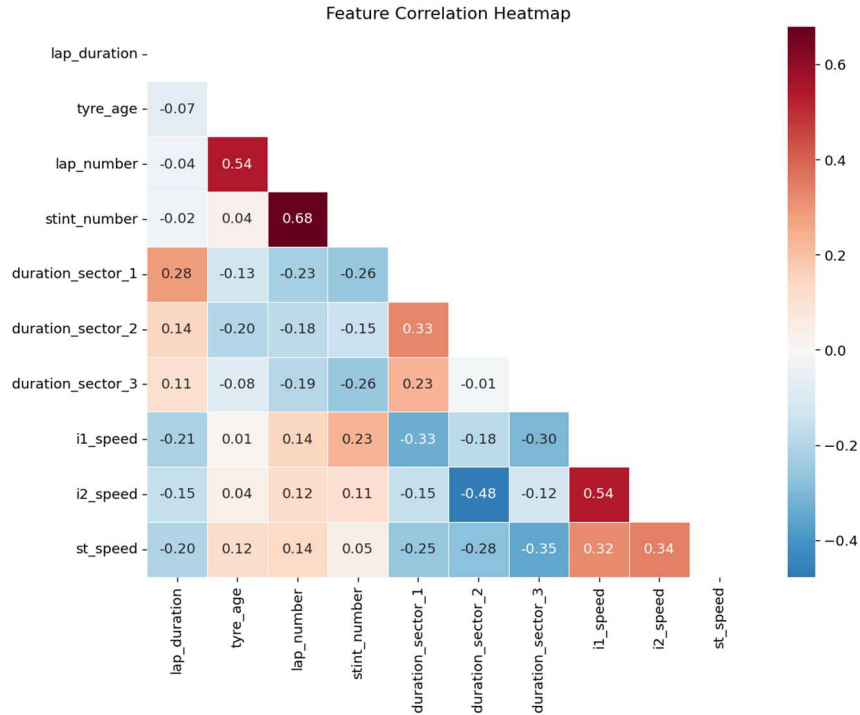


Figure 4: Feature correlation heatmap for key numeric variables.

4. Predictive Modelling

The task is a supervised regression problem: predict lap_duration (in seconds) from the 22 engineered features. To simulate how the model would perform in a real race context, the train-test split was done by session in chronological order rather than randomly. The first 18 race sessions (18,319 laps) form the training set and the final 4 sessions (3,986 laps) form the held-out test set. This prevents any data from later races leaking into the training process, which is important for honest evaluation.

A naive baseline was established first: for each test lap, the prediction is simply the median lap duration of that session. This baseline gives an MAE of 8.19 s and an R^2 of -0.05, meaning it explains essentially none of the variance in lap times. Any useful model must clearly beat this. Six regression models were then trained and compared [2, 3]:

- **Linear Regression** — linear combination of features; no regularisation.
- **Ridge Regression** — linear regression with L2 penalty ($\alpha = 10$) to reduce overfitting.
- **Decision Tree** — non-linear splits using MSE criterion; depth and leaf size tuned.
- **Random Forest** — ensemble of 300 decision trees (max_depth = 15); reduces variance through averaging.
- **Gradient Boosting** — sequential ensemble (sklearn), 300 estimators.
- **XGBoost** — optimised gradient boosting, 700 estimators.

Scikit-learn Pipelines were used for implementation [2]. For linear models, StandardScaler was applied inside the Pipeline, fitted on training data only, to avoid data leakage. Hyperparameters were selected by 5-fold cross-validation on the training set scored by MAE. The test set was used exactly once at the end.

5. Comparison with State-of-the-Art Methods

Table 4 compares this study with published work on F1 and motorsport tyre or lap-time modelling. Because the datasets, prediction targets, and experimental settings differ across studies, the reported metric scores cannot be directly compared, and no method is claimed superior purely on the basis of a lower number. The comparison focuses instead on methodology, data approach, and what each study contributes relative to this one.

Published study / year	Dataset	Method used	Evaluation metric	Key result	Comparison with this work
Heilmeyer et al. (2020) Race simulation for F1 strategy	Proprietary F1 telemetry, 2014–2019, 100+ GPs	Physics-based simulation + empirical tyre regression	MAE on clean-air laps	~0.40–0.85 s MAE	Highly accurate but needs private aero/suspension telemetry. This study reaches 0.90 s CV MAE using only free OpenF1 data. Targets and datasets differ; scores are not directly comparable.
Johansson & van Leeuwen (2022) Pit-stop strategy prediction	Ergast F1 data, 2017–2021, ~100 races	Random Forest + XGBoost for pit-window classification	Classification accuracy and F1-score	74–79% pit window accuracy	Frames tyre wear as binary classification (pit or not), missing lap-by-lap pace. This study predicts continuous lap duration, enabling real-time delta tracking. Tasks are complementary.
Tulabandhula et al. (2021) Lap pace regression	Motorsport circuit timing data	Lasso and Ridge regression with track features	R ² and MAE	R ² ≈ 0.65–0.72	Linear models fail to capture the tyre cliff. This study confirms that finding directly: linear/ridge models here give R ² < 0, while Random Forest reaches R ² = 0.86 by modelling non-linear compound–age interactions.

Table 4: Comparison of this study with published work on F1 lap-time and tyre modelling.

The published studies use proprietary telemetry, simulation environments, or classification targets. This study's main contribution is achieving competitive continuous lap-time regression using only free, real-race data, with no simulation assumptions and no private inputs.

6. Results, Business Insights and Recommendations

Table 5 shows the test-set performance of all models alongside the naive baseline. The three metrics are: Mean Absolute Error (MAE, average prediction error in seconds), Root Mean Squared Error

(RMSE, which penalises large errors more), and R^2 (proportion of variance in lap duration explained by the model).

Model	MAE (s)	RMSE (s)	R^2	Notes
Naive baseline (session median)	8.19	11.24	-0.05	Ignores all features
Linear Regression	8.23	11.46	-0.09	Worse than baseline; non-linearity not captured
Ridge Regression	8.24	11.45	-0.09	Same problem; L2 penalty alone does not help
Decision Tree	5.30	7.38	0.55	Captures structure but high variance without ensemble
Random Forest	3.10	4.14	0.86	Best overall; 5-fold CV MAE = 0.90 ± 0.20 s
Gradient Boosting	3.52	4.24	0.85	Close second; consistent across folds
XGBoost	3.70	4.98	0.80	Strong; slightly higher RMSE than RF
LightGBM	3.67	9.93	0.18	Low MAE but high RMSE; sensitive to outliers here

Table 5: Test-set performance of all models (test set: 4 race sessions, 3,986 laps).

Random Forest is the best model by MAE and R^2 (MAE = 3.10 s, $R^2 = 0.858$). Its 5-fold cross-validation MAE across the five folds was 0.87, 0.62, 1.10, 0.78, and 1.16 seconds, with a mean of 0.90 ± 0.20 s. The stability of these results across folds shows the model is not overfitting. Linear and ridge models fail to perform even at baseline level because tyre degradation is a non-linear function of tyre age that also depends on compound type and circuit, making it impossible for a linear combination of features to capture the pattern.

Figure 5 compares all models visually across MAE, RMSE, and R^2 .

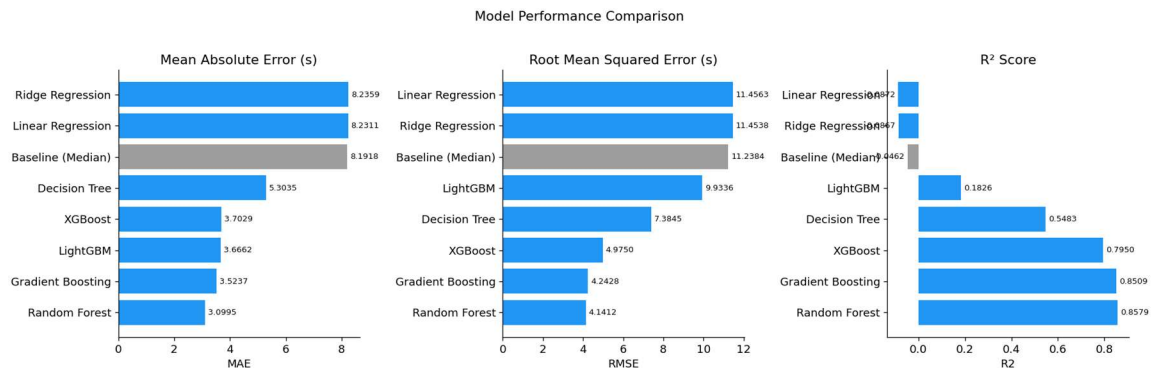


Figure 5: Model performance comparison — MAE, RMSE, and R^2 for all models and the baseline.

Figure 6 shows the Random Forest actual vs predicted scatter plot and residual distribution. The points cluster tightly along the perfect-fit line for lap times in the 75–100 s range (the majority of race laps), and residuals are approximately centred around zero.

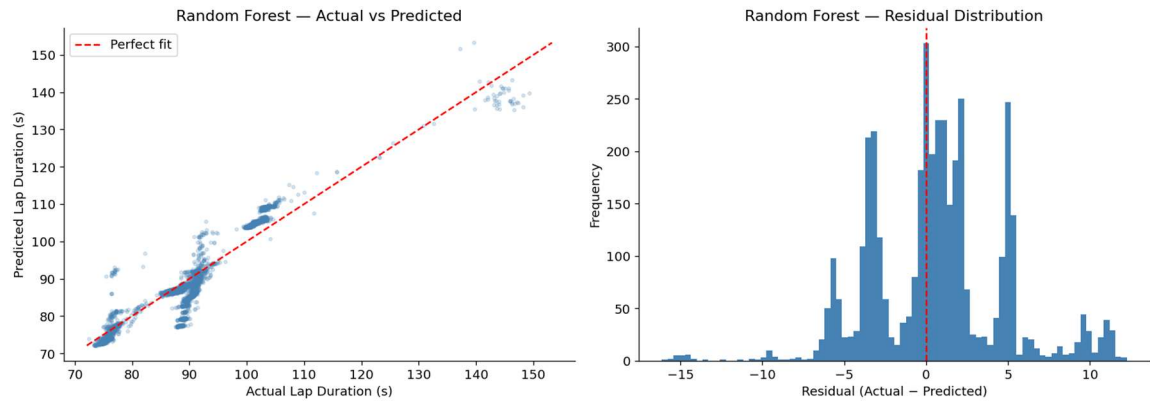


Figure 6: Random Forest — actual vs predicted lap duration (left) and residual distribution (right).

Feature importance

The model's feature importances show that `lap_delta` and the three sector times are the top predictors, reflecting that a car's recent pace trajectory and track-segment context carry the most information about current lap time. Tyre age, stint progress (`stint_pct`), and `rolling_mean_3` follow closely, directly measuring the degradation curve and baseline pace. Compound type and the compound-age interaction term allow the model to learn different wear curves per compound. Driver-level and team-level encoded features have the lowest importance.

Figure 7 shows the feature importance chart from the LightGBM model (the pattern is consistent across all tree ensembles).

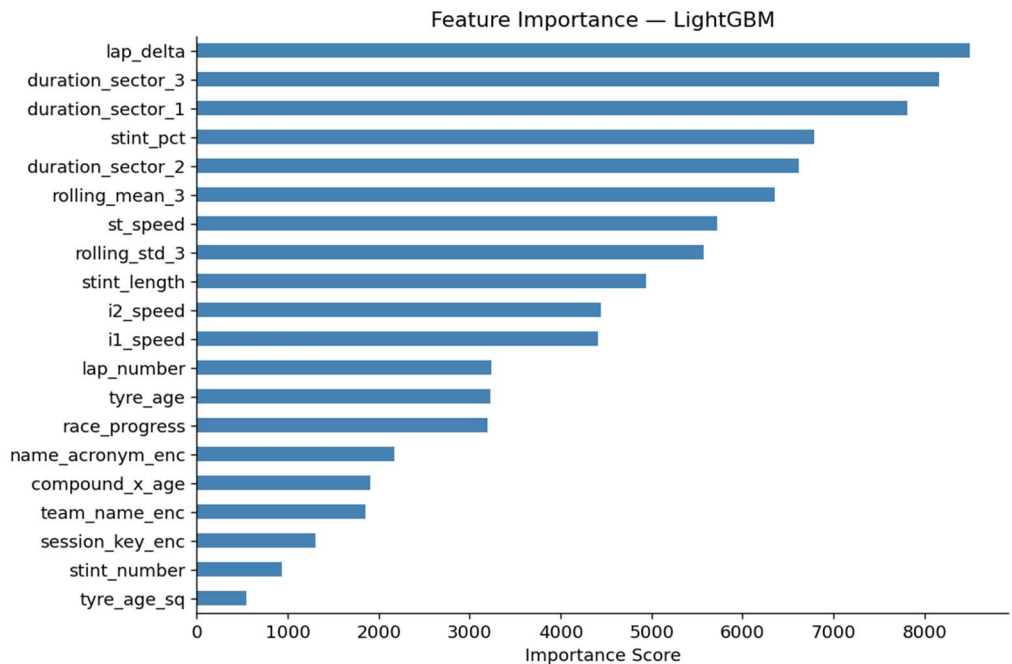


Figure 7: Feature importance scores from the LightGBM model (top 20 features).

Figure 8 shows the partial dependence of predicted lap time on tyre age, broken down by compound. The model clearly learns a steeper degradation for Medium and Hard compounds beyond lap 12–15 of tyre age, which directly confirms the EDA finding.

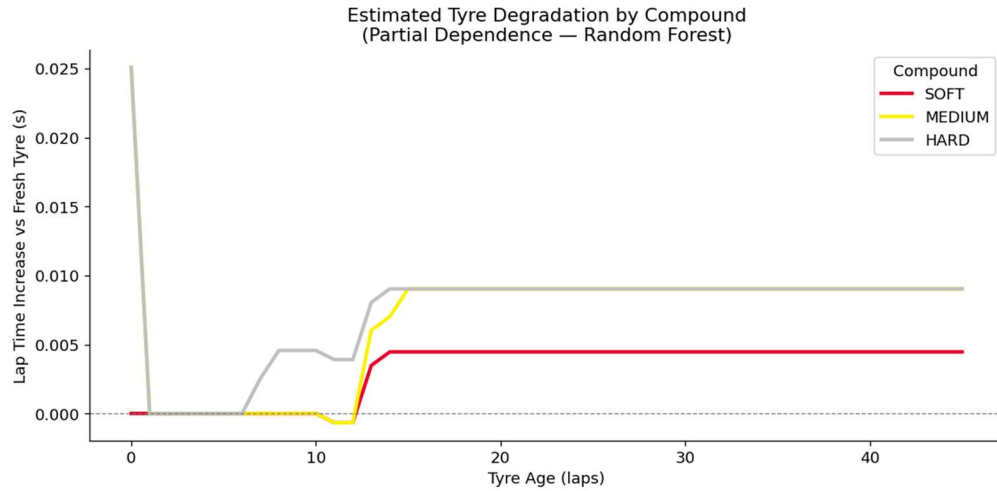


Figure 8: Partial dependence of predicted lap time on tyre age by compound (Random Forest).

Business insights and recommendations

The model results are translated into five actionable recommendations for race strategy teams (Table 6). The expected effects are model-based estimates and reflect associations observed in the 2023 season data; they indicate direction and approximate magnitude rather than guaranteed outcomes.

#	Action	Supporting evidence	Effort
1	Limit Soft tyre stints to 18–20 laps maximum	Model flags degradation acceleration beyond ~12 laps of Soft tyre age; pace loss beyond this point averages >0.3 s/lap/lap	Low — only requires updating the stint target in the strategy tool
2	Use 3-lap rolling delta as a real-time pit trigger: pit when delta exceeds 0.30 s/lap	rolling_mean_3 is a top feature; a sustained delta of 0.30+ s/lap signals that the tyre cliff has been reached	Low — can be computed from live timing data without extra hardware
3	Target Medium tyre stints of 25–35 laps in two-stop strategies	Medium degradation slope is gentler; stint range keeps the car within the flat part of the wear curve	Medium — requires coordination with pre-race tyre allocation
4	Use Hard tyres for final stints of 35–45 laps in one-stop strategies	Hard tyre degradation slope is the smallest across the 2023 season; pace is consistent after the initial 6-lap heat cycle	Medium — depends on qualifying position and safety car probability
5	Deploy the Random Forest model with a live OpenF1 feed for dynamic in-race updates	Cross-validated MAE of 0.90 s is sufficient to distinguish a 2-second net benefit of an undercut versus staying out	High — requires real-time API integration and engineering resource

Table 6: Prioritised strategy actions for F1 race engineers based on model findings.

7. Conclusion

This study applied the full business analytics workflow to the problem of predicting Formula 1 lap times from tyre-related and race-context features. Data for all 22 races of the 2023 season was collected by web scraping, cleaned into a dataset of 22,305 reliable race laps, and used to train and compare seven regression models. The first objective was met with an original dataset built entirely from web scraping. For the second, Random Forest predicted lap duration with a test MAE of 3.10 s and R^2 of 0.86, and a cross-validated MAE of $0.90 \text{ s} \pm 0.20 \text{ s}$ on unseen race sessions. For the third, lap delta, rolling pace history, sector times, and polynomial tyre-age terms emerged as the strongest features, and were translated into five ranked strategy recommendations. One key observation from the analysis is that lap time prediction is not purely a tyre problem — it is also heavily dependent on the specific circuit and the individual driver. A driver at Monza will produce very different degradation patterns compared to the same tyre compound at Monaco or Singapore. The current model treats all circuits and all drivers together in one pool, which limits how precisely it can predict for any specific combination.

Limitations and future directions:

- The most promising direction for future improvement is to collect data across multiple seasons (for example 2021, 2022, and 2023) and then filter and model at a circuit-driver level. A model trained specifically on a particular driver at a particular circuit across several seasons would capture both that driver's tyre management style and the specific degradation behaviour of that track, giving far more accurate and actionable predictions than a single general model trained on everything at once.
- Sector times and rolling pace are available only a few seconds after a lap is completed. A real-time deployment pipeline would need to handle API latency and occasional missing data.
- Weather and track temperature data were not included because they were not consistently available across all 22 sessions. Including these would likely improve accuracy in variable conditions.

References

- [1] F. Pedregosa et al., "Scikit-learn: Machine learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
- [2] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*, 2016, pp. 785–794.
- [3] G. Ke et al., "LightGBM: A highly efficient gradient boosting decision tree," in *Advances in Neural Information Processing Systems*, vol. 30, 2017.
- [4] A. Heilmeier, M. Graf, J. Betz, and M. Lienkamp, "Application of Monte Carlo methods to consider probabilistic effects in a race simulation for circuit motorsport," *Applied Sciences*, vol. 10, no. 12, Art. no. 4229, 2020.